**Protocol fixed** (input, splits, operating point, seeds); approved and tagged.

**Development on train/validation**: whole-card, regularised and tile models; tile model selected; 5 seeds.

**Calibration and A4 fusion** fitted on validation; parameters committed; signed by all authors.

**Test split scored once** after a freeze tag; sealed scores hash-checked.

**A5 designed** from test findings; design recorded before fitting; reported as post-hoc.

**Fresh set pre-registered**, generated, leakage-checked and evaluated once (confirmatory test of A5).